

Wiser Summer Program – 2026

Lecture Slides

Session 1

by Scott Crowder

(Jul 8, 2026)

Path to Useful Quantum Computing and the Role of Quantum & AI in Optimization

Scott Crowder
Vice President, IBM Quantum Adoption



00

Intro to Quantum Computing

01

Path to Useful Quantum Computing

02

Optimization Landscape

03

Quantum for Optimization

04

Recent AI for Optimization Research

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Quantum computing is based on a new information science that leverages quantum mechanics

Based on insight that Nature is beyond binary arithmetic and we can leverage quantum mechanics to create a new information science



A quantum bit or **qubit** is a controllable quantum object that is the unit of information



A **quantum circuit** is a set of quantum gate operations on qubits and is the unit of computation

Creates new ways to solve problems that are impossible for digital computers

That has potential to create large business value at tech maturity

"Quantum computing has the potential to create up to \$850B in value at tech maturity." —BCG

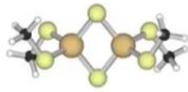
Sources: BCG, "The Quantum Leap: Navigating the Future of Computing," 2025; "The Long-Term Forecast for Quantum Computing Still Looks Bright," 2024.

© 2025 IBM Corporation 4:14 / 52:04

Pause, CC, Settings icons

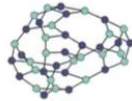
Key quantum computational areas

Hamiltonian simulation



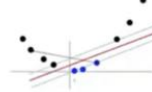
Quantum computers solve Hamiltonian simulation by directly evolving quantum states under the same physical laws described by the Hamiltonian, enabling efficient modeling of complex quantum systems.

Optimization



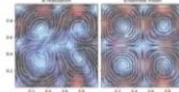
Quantum computers solve optimization problems by encoding them into quadratic or higher-order unconstrained binary optimization forms, then using quantum amplification to search for the optimal solution.

Machine learning



Quantum computers can map data into high-dimensional Hilbert spaces—defining quantum kernels that capture complex group symmetries and correlations beyond classical computation, enabling powerful classifiers.

Differential equations



Quantum computers can efficiently model strongly nonlinear and noisy classical dynamics in polynomial time for fixed noise rates—by lifting them into a Hilbert space where evolution becomes efficient.

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6:34 / 52:04

Advancement of quantum computing

2016

Quantum is online

Quantum devices on the cloud, opening the path for quantum developers

2023

Quantum utility

Quantum computers run quantum programs beyond classical

2026

Quantum is real

Provable scientific quantum advantage

Quantum applied to real use cases, simulating problems of interest

2029

Fault tolerance

>100 million operations with broad resistance to failures

2035

Cryptographic relevance

Expected capabilities to run Shor's algorithm and break today's encryption schemes

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Where is quantum computing today?

Quantum Computers are real and available as computational tools for algorithmic discovery

Our state-of-the-art quantum computers can run programs with 5000+ operations, which is too complex to run on world's largest classical supercomputer

IBM Quantum Platform provides continuous access to these 100+ qubit computers to clients around the world

IBM has deployed 90+ devices since 2016 including 30+ of over 100 qubits

9 quantum computers on-site



2 quantum data centers



- ✓ Average 97% availability across fleet
- ✓ 99% availability of at least one Heron R2+
- ✓ Job success rate >99.4%

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10

Where is quantum computing today?

Software stack based on open source has emerged

Qiskit

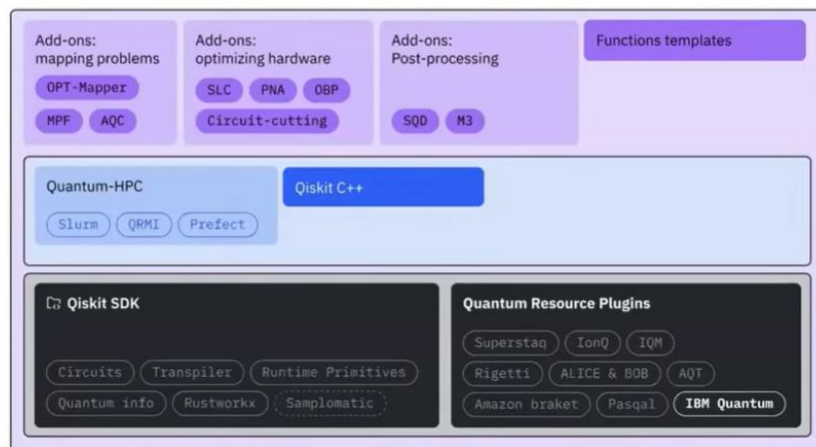
Open ecosystem for quantum computing

69% Quantum-native developers using Qiskit
(2025 Unitary Foundation Survey)

Qiskit for:
■ Algorithm research

Qiskit for:
■ High Performance Computing

Qiskit for:
■ Quantum information science research



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Availability of quantum technology driving growth of innovation in quantum algorithm and applications

Community taking advantage of learning opportunities

40K+

Engaged learners in past 12 months

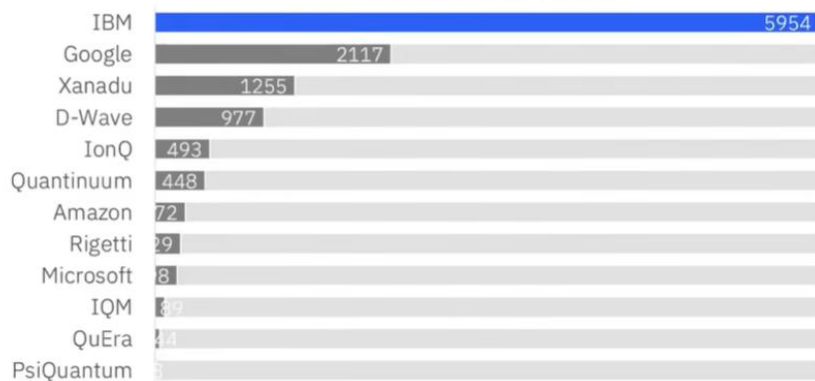
15K+

Total badges earned

32K+

Students attended Qiskit Fall Fest in 2025

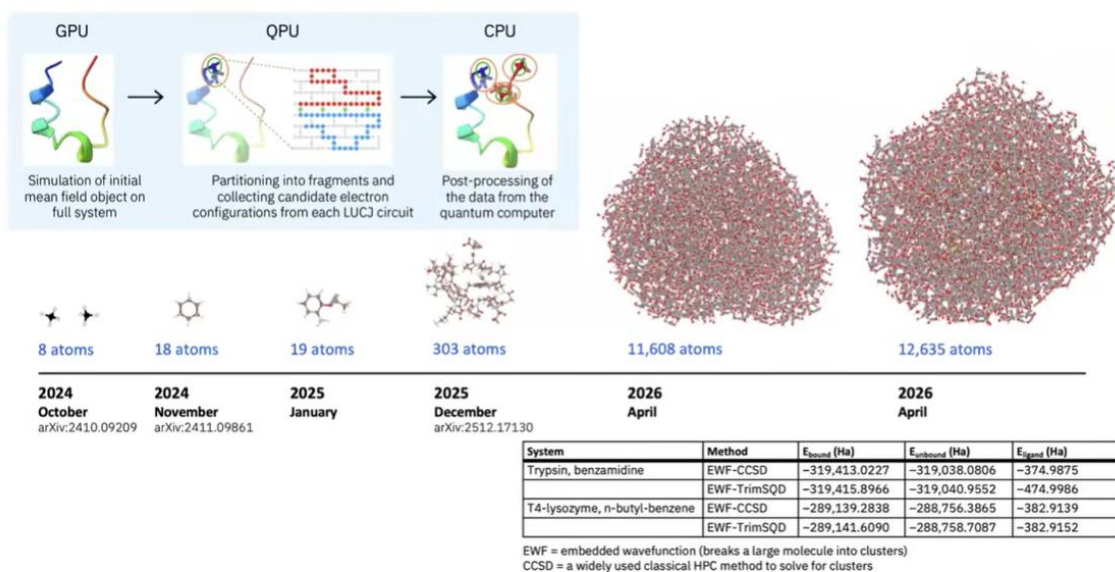
Number of documents on arXiv referencing vendor's quantum hardware and/or software



Data is current as of 4/22/26. The data was obtained by (1) collecting all arXiv papers mentioning quantum computing related phrases; (2) cleaning up the corresponding LaTeX files and removing bibliographies/citations/formatting; and (3) searching for SW and service keywords found in the [2024 Unitary Foundation Survey](#) and publicly known quantum HW names, with the constraint that an action word (e.g., "execute," "run") is required to exist in the same sentence with the keyword(s).
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Pushing the scale of chemistry applications

Cleveland Clinic | IBM Quantum



Industry leaders are building skills and exploring use for business-relevant applications

Algorithms connect computational capabilities to industry applications

100+

Completed prototypes with industry clients



Exploring material degradation caused by radicals

arXiv:2508.08229

Hamiltonian simulation



Predicting mRNA secondary structures with quantum optimization techniques

arXiv:2405.20328

Optimization



Enhancing trade execution strategies for corporate bonds

arXiv:2509.17715

QML



Portfolio optimization for fixed income assets

arXiv:2508.13557

Optimization



Decentralized energy system optimization

arXiv:2407.20802
arXiv:2412.07915
arXiv:2509.10657

Optimization

QML



Simulating supramolecular interactions in pharmaceutical systems

Communications Physics 8, 396 (2025)

Hamiltonian simulation



Simulating advection-diffusion and similar partial differential equations

arXiv:2508.16728

PDEs

Two key prerequisites for bringing useful quantum computing to the world

Developing a quantum platform that scales beyond classical computation



Activating a network of enterprises and academic institutions to advance algorithm discovery



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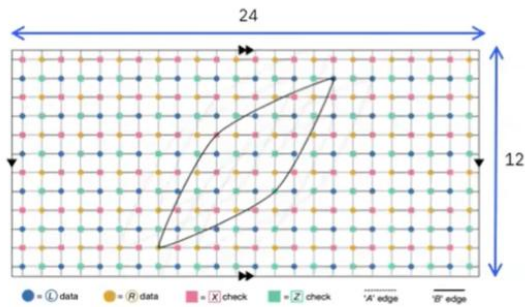
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Why do we have confidence in 2029 to deliver fault tolerant quantum computing?

- We have developed a more efficient error correction code and demonstrated the ability to run the decoder software on relatively low-cost electronics

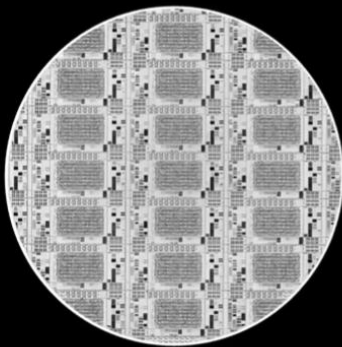


The gross code is 10x more efficient than the surface code

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Building the first large-scale, fault-tolerant quantum computer



IBM Quantum Loon

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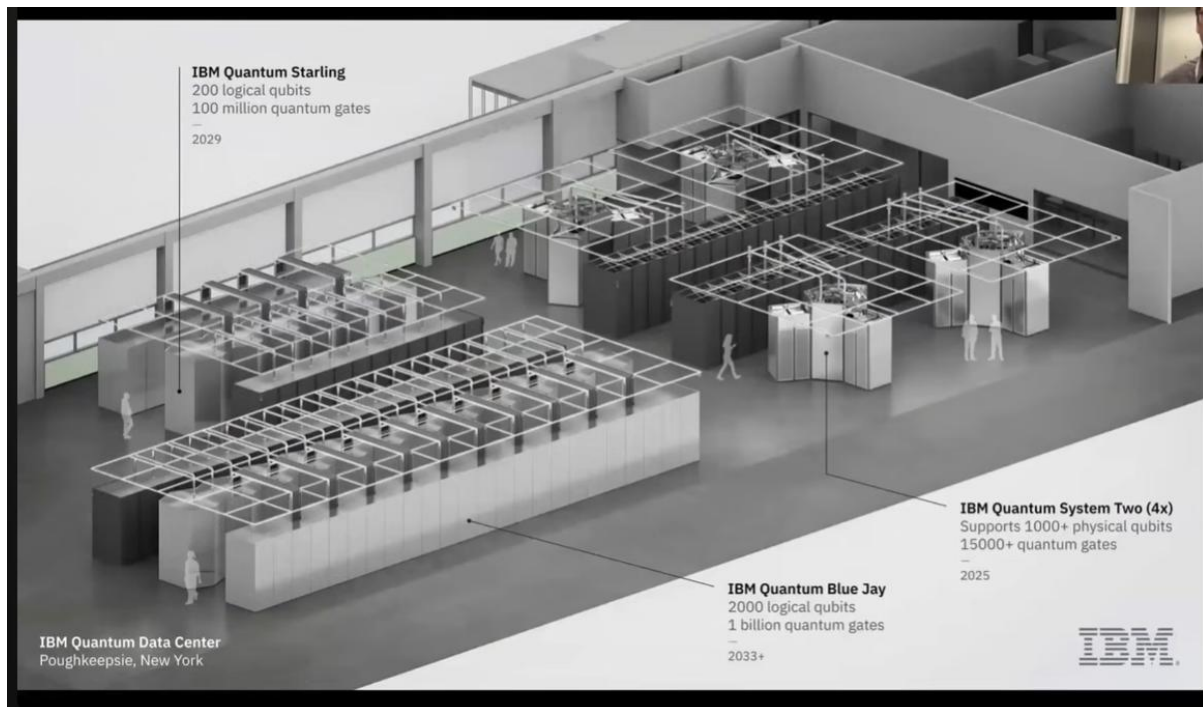
2025 2026 2027 2028

Loon Demonstrate c-couplers and next-generation packaging for FTQC	Kookaburra Demonstrate a complete module consisting of a logical processing unit and quantum memory	Cockatoo Demonstrate entanglement of modules using a universal adapter	Starling Demonstrate multiple modules and magic state distillation
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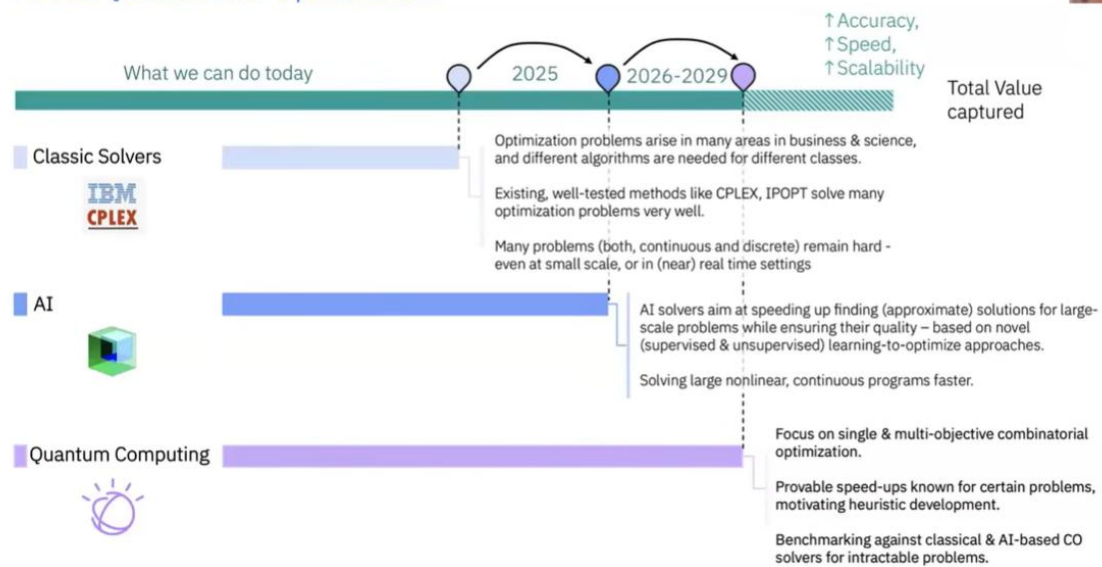
2029

2033+

Starling (100M) Fault-tolerant 100M gates 200 qubits	Blue Jay (1B) Fault-tolerant 1B gates 2000 qubits
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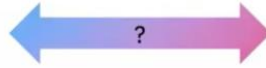
AI and Quantum for Optimization



Quantum optimization

Probably one of the most **misunderstood** areas in quantum computing

“Quantum computers can
evaluate all possible
solutions at the same time”



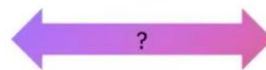
“Quantum computers can
achieve at most a
quadratic speed-up”

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Quantum optimization

Probably one of the most **misunderstood** areas in quantum computing

“Quantum computers can
evaluate all possible
solutions at the same time”



“Quantum computers can
achieve at most a
quadratic speed-up”

This statement usually refers to difficult problems (NP-hard) where the runtime of provably exact classical algorithms in general (= worst case) scales exponentially with the problem size
→ quadratic speed-up: $O(2^n) \rightarrow O(2^{n/2})$

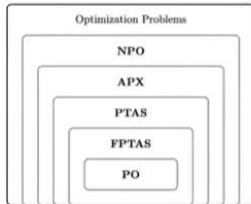
If this would be the end of the story, then, neither classical nor quantum algorithms would be very useful for (combinatorial) optimization!

Luckily, the situation is much more nuanced than that!

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Quantum Advantage in Optimization?

What's **known** or **not ruled out** by complexity theory.



Provably Exact Algorithms

Guaranteed to return global optimal solutions

NP-hard problems (MAXCUT, TSP, ...)

In general (incl. worst case), at most **quadratic speedup**, e.g., via Grover search (possibly sub-routine in classical algorithm).

NP-intermediate problems (Factoring, Graph Isomorphism, ...)

Optimization problems that can be mapped to factoring and can be solved with **exponential speedups** via Shor's algorithm (unlikely to be practically relevant)

P problems (Convex optimization like LP/SDP, ...)

Exponential speedups are possible! If input/output can be handled efficiently, e.g., via functional descriptions of problems (need to make sure these cannot be exploited classically too!)

Provably Approximate Algorithms

Return a solutions with a priori performance guarantee

APX (approximable) problems with tight inapproximability bounds (MAXCUT: 87.8%)

Possible runtime advantages as for problems in P but with same performance guarantees.

APX (approximable) problems with loose inapproximability bounds (Metric TSP: 150% > 101.3%)

Possible runtime advantages as for problems in P but possibly with **better approximation ratios!**

"Decoded Quantum Interferometry" (Jordan et al. arXiv:2408.08292) is a recent example of an **exponential speedup** with respect to best known classical algorithms for the "Optimal Polynomial Intersection" problem.

PTAS (Knapsack, Euclidean TSP, ...)

Possible runtime advantages as for problems in P.

Heuristic Algorithms

Often return good solutions for some instances, no guarantees

No a priori guarantees.
Need to try and see how things work.
Often heuristics work very well in practise.

Some heuristics provide a posteriori bounds on the optimality gap, which can result in provably optimal solutions for instances of NP-hard problems.

→ Most quantum optimization algorithms are heuristics (particularly with noise)!
→ We need to run them and find out how well they perform!

Example of the the performance of a classical heuristic:

The "World TSP", a problem with 1.9M cities, was solved in 2021 to 0.0471% optimality!
<https://www.math.uwaterloo.ca/tsp/world>

Disruptive quantum advantage in optimization is possible!

*Many quantum optimization algorithms are / will be heuristics (like classically)
→ We need to test them and learn where they outperform classical algorithms!*

Combinatorial optimization

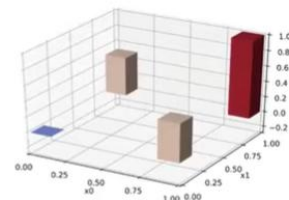
Most near-term quantum optimization algorithms consider combinatorial optimization...

- Usually no I/O bottleneck (search space is exponentially larger than description of problem & solution)
- Even small instances can be very challenging (classically)!

...and, in particular, quadratic/higher-order unconstrained binary optimization (QUBO/HUBO)

- QUBO/HUBO has natural mapping to ground state problems for Ising Hamiltonians:

$$\min_{x \in \{0,1\}^n} x^T Q x + b^T x \quad \longrightarrow \quad \min_{|\psi\rangle} \langle \psi | H_C | \psi \rangle$$



$$\min_{x \in \{0,1\}^2} x_0^2 + x_1^2 - (x_0 + x_1)/2$$

$$\begin{aligned} & \downarrow \\ & -\frac{1}{4}(Z \otimes \mathbb{I} + \mathbb{I} \otimes Z) + \frac{1}{2} \\ & = \begin{pmatrix} 0 & 1/2 & 1/2 & 1 \end{pmatrix} \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix} \end{aligned}$$

Quantum approximate optimization algorithm (QAOA)

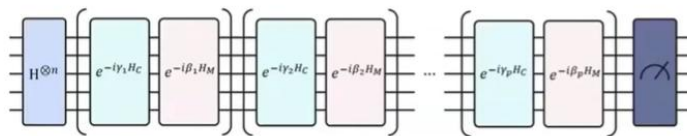
Inspired from quantum adiabatic algorithm:

$$H(t) = (1 - t)H_M + tH_C$$

Discretize the time-evolution into p rounds and optimize the parameters γ_i and β_i :

$$\min_{\beta, \gamma} \langle \psi(\beta, \gamma) | \hat{H}_C | \psi(\beta, \gamma) \rangle$$

Theoretical performance guarantees in certain cases and in the limit $p \rightarrow \infty$.



QAOA circuit with p rounds

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Farhi et al., arXiv:1411.4028 (2014)

Challenges for quantum advantage with QAOA

Scale

Qubit count/connectivity

- Current devices are limited to $O(100)$ qubits, which usually translates to $O(100)$ binary variables.
- Devices will scale to larger numbers of qubits and higher connectivity.

Quality

Noise

- Hardware noise limits the practical depth of circuits.
- Particularly for dense QUBOs, QAOA circuits can become rather deep.
- Devices will become more reliable, ultimately, error correction will suppress noise exponentially.

Speed

Sampling/training

- Need many samples to extract high-quality results.
- Variational training introduces significant computational overhead.
- Fastest quantum computers (based on superconducting qubits) achieve sampling rates of $O(10\text{kHz})$.

Performance

Heuristics

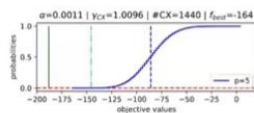
- In general, QAOA is a heuristic, i.e., its performance is problem instance dependent and not known upfront.
- Rigorous benchmarking is crucial to assess the performance and track progress towards quantum advantage.

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Increasing number of utility-scale demonstrations* on IBM quantum hardware charting path toward quantum advantage in optimization!

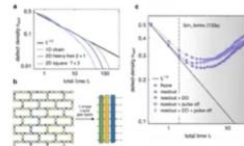
* Often related to QAOA or variants of it!

Provable bounds for noise-free expectation values computed from noisy samples



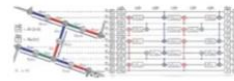
What: up to 127 qubits on HW native 3rd order with $p = 5$.
Barron et al. (2023)
arXiv:2312.00733

Benchmarking digital quantum simulations and optimization using quantum critical dynamics



What: up to 133 qubits on HW native Ising problems.
Miessen et al. (2024)
arXiv:2404.08053

Quantum Annealing vs. QAOA: 127 Qubit Higher-Order Ising Problems on NISQ Computers



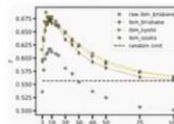
What: 127 qubits on HW native 3rd order problems with $p = 2$.
Pelofske et al. (2023)
ISC High Performance 2023

Large-scale QAOA on non-graphs with machine learning noise mitigation



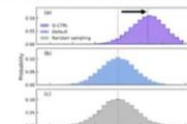
What: up to 40 qubits on RR3 graphs and depth $p = 2$.
Sack & Egger (2024)
Phys. Rev. Research 6, 2023

Towards a universal QAOA protocol: Evidence of quantum advantage in solving combinatorial optimization problems.



What: up to 105 qubits on HW native with p up to 100.
Montanez-Barrera et al. (2024)
arxiv:2405.09169

Q. opt. using a 127-qubit gate-model IBM quantum computer can outperform quantum annealers for nontrivial binary opt. problems



What: up to 127 qubits on HW with error suppression.
Sachdeva et al. (2024)
arxiv:2406.01743

Qiskit Functions for quantum optimization

Cloud services offered on top of IBM Quantum Platform



Iskay Quantum Optimizer

Request a free trial

Read the docs

Submit your results via the third-party site are subject to the provider's terms and conditions.

Features and benefits

- High solution quality**
Solve complex optimization problems of up to 156 qubits with high solution quality.
- Short time-to-solution**
Obtaining a solution faster due to non-variational nature of the algorithm.
- Automated workflow**
Easy to use quantum optimizer lowering the bar to use quantum computing for business relevant optimization problems.

Related papers

Scientific papers related to or supporting this function.

Bias-Field Digitized Counterdiabatic Quantum Algorithms for Higher-Order Binary Optimization
Romero, Vissani et al. (2401.04477 (2023))

Bias-Field Digitized Counterdiabatic Quantum Optimization
Cabrero, Dával et al. (2405.13899 (2023))

Optimization Solver

Request a free trial

Read the docs

Submit your results via the third-party site are subject to the provider's terms and conditions.

Features and benefits

- Achieve quality results at utility scale**
Successfully solve computationally challenging problems with over 100 qubits.
- Build quantum solutions with ease**
Leverage a fully-packaged, hybrid quantum-classical solver without any quantum expertise.
- Save time and resources**
Converge on the right answer faster with fewer iterations and fewer shots.

Related tutorials

Learn how to apply this technology to your workflow.

Beginner
Higher-order binary optimization with Q-CTRL's Optimization Solver

Related papers

Scientific papers related to or supporting this function.

Quantum Optimization Benchmarking Library (QOBLIB)

Quantum Optimization Benchmarking Library
The Intractable Decathlon

Thorsten Koch ^{*1,2}, David E. Bernal Neira³, Ying Chen⁴, Giorgio Cortiana⁵,
Daniel J. Egger⁶, Raoul Heese⁷, Narendra N. Hegade⁸, Alejandro Gomez
Cadavid^{9,9}, Rhea Huang¹⁰, Toshiaki Ito¹¹, Thomas Kleinert¹², Pedro Maciel
Xavier¹³, Naeimeh Mohseni³, Jhon A. Montanez-Barrera¹⁴, Koji Nakano¹⁵,
Giacomo Nannicini¹⁰, Corey O'Meara³, Justin Pauckert¹⁶, Manuel Proissl⁸, Anurag
Ramesh¹, Maximilian Schicker¹, Noriaki Shimada¹¹, Mitsuharu Takeori¹¹, Victor
Valls¹⁷, David Van Bulck¹⁸, Stefan Woerner¹⁸, and Christa Zoufal¹⁸

¹Zuse Institute Berlin
²Technische Universität Berlin
³Davidson School of Chemical Engineering, Purdue University
⁴National University of Singapore
⁵E.ON Digital Technology GmbH
⁶IBM Quantum, IBM Research Europe - Zurich
⁷NTT Data
⁸Rigetti Quantum GmbH
⁹University of the Balearic Islands UPF/IBU
¹⁰University of Southern California
¹¹IBM Quantum, IBM Research Tokyo
¹²Quantagena
¹³Federal University of Rio de Janeiro
¹⁴Forschungszentrum Jülich
¹⁵Hiroshima University
¹⁶T-Systems International GmbH
¹⁷IBM Quantum, IBM Research Europe - Dublin
¹⁸Ghent University

August 29, 2025

arXiv:2504.03832

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QOBLIB - Quantum Optimization Benchmarking Library

QOBLIB - Quantum Optimization Benchmarking Library

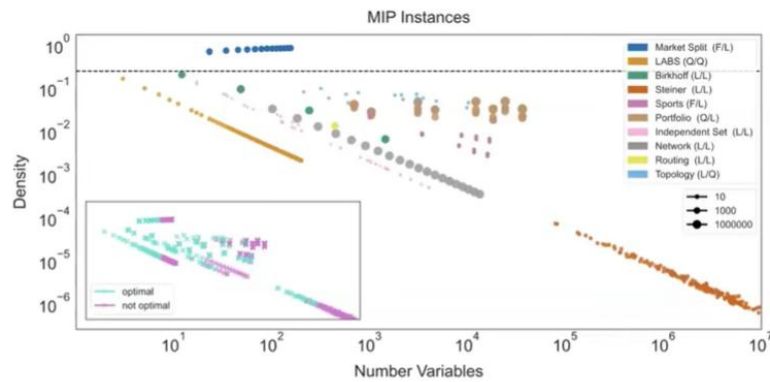
Find file Edit Code

49723ae 1 hour ago

Name	Last commit	Last update
08-mapping	Fix Contribution and terminology	1 day ago
09-lab	08 README.md	16 hours ago
09-birkhoff	Birkhoff Files	23 hours ago
09-stein	Fix Contribution and terminology	1 day ago
09-sports	Initial commit	2 days ago
09-portfolio	Initial commit	2 days ago
09-independent	Remove unnecessary files	22 hours ago
09-network	Initial commit	2 days ago
09-routing	Fix Contribution and terminology	1 day ago
09-topology	Fix Contribution and terminology	1 day ago
09-misc	Fix Contribution and terminology	1 day ago
gitignore	Initial commit	2 days ago
CONTRIBUTING	Fix Contribution and terminology	1 day ago
README	Fix Contribution and terminology	1 day ago

<https://git.zib.de/qopt/qoblib-quantum-optimization-benchmarking-library>

Quantum Optimization Benchmarking Library (QOBLIB)



- QOBLIB is an open-source community-driven benchmarking effort.
- Together with the community, we want to define how quantum advantage in optimization may look like.
- We invite everybody to contribute quantum and classical results to track progress and advance toward quantum advantage!

→ <https://github.com/ZIB-AOPT/QOBLIB>

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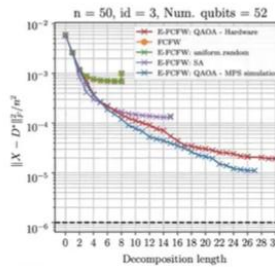
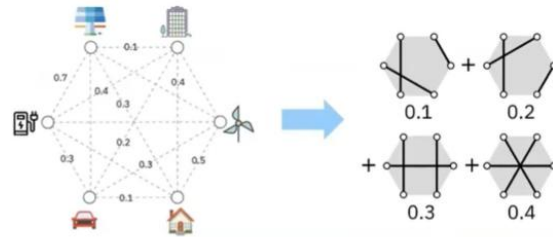
Extension of the QOBLIB problem Minimum Birkhoff Decomposition

From bipartite to general graphs

Goal: Decompose a weighted graph using the minimum number of graph matchings

Extension is motivated by peer-to-peer energy exchange:

- Nodes want to buy/sell energy
- Edges represent the normalized amount of energy two nodes want to exchange
- Energy exchange is constrained in practice: only two nodes can exchange energy at a time
- Goal: Minimize the number of transactions, which affect the total energy exchanged per unit of time



Hardware experiments with 52–111 qubits (plot with 52 qubits)

Algorithm using QAOA shows better performance than a greedy classical approach. In some cases, QAOA finds a decomposition smaller than the best-known solution!

Pennington et al., arXiv:2509.10657 (2025)

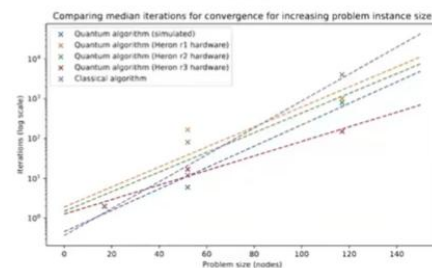
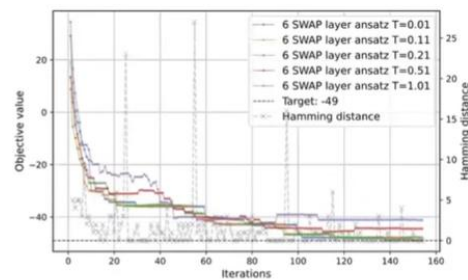
Quantum-enhanced Markov chain Monte Carlo (QeMCMC) for combinatorial optimization

Marshall et al., arXiv:2602.06171 (2026)

- Novel algorithm combines QeMCMC with warm-starting QAOA and parallel tempering to tackle constrained optimization
- Based on promising results from QeMCMC paper on Boltzmann sampling with up to quartic speedup over classical MCMC [1].
- Tested on Maximum Independent Set (MIS) up to 117 nodes/qubits, demonstrating promising scaling on IBM quantum hardware compared to classical MCMC and MPS simulation.
- Instances extracted from Quantum Optimization Benchmarking Library, QOBLIB [2], to help benchmark the progress to quantum advantage in optimization.

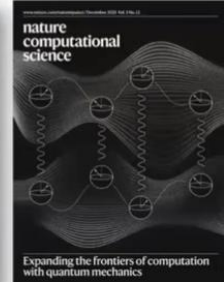
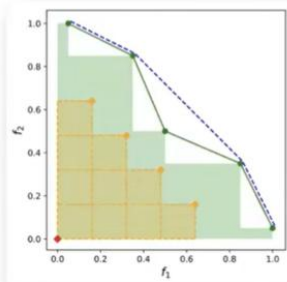
[1] Layden et al. Quantum-enhanced Markov chain Monte Carlo. Nature (2023).

[2] Koch et al. Quantum Optimization Benchmarking Library - The Intractable Decathlon (2025).



Charting new paths to quantum advantage: Multi-Objective Optimization (MOO)

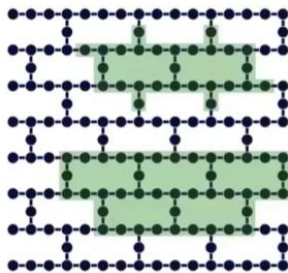
- The goal of MOO is to find the Pareto front, i.e., **the set optimal trade-offs of multiple competing objective functions** (e.g. risk, return, diversification, service level, cost, sustainability, ...), relevant, e.g., in Finance or Logistics.
- MOO is **significantly more difficult than single-objective optimization**. Gets intractable (to solve exactly) with increasing number of objective functions even at relatively small problem sizes!
- We have shown that **quantum can help to explore the Pareto front faster** than classical algorithms.



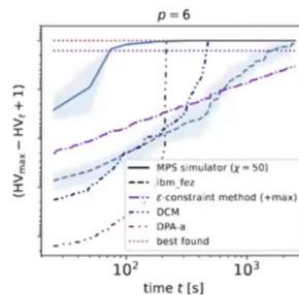
A&A Deep Dive 2026

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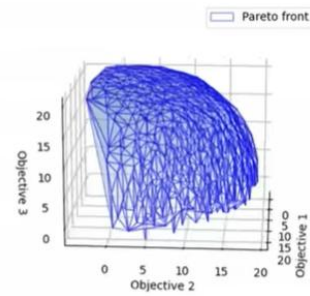
A promising candidate for quantum advantage in optimization...



Topology of the `ibm_fez` Heron device



Hypervolume over time



Optimal Pareto front computed by `ibm_fez`

Quantum optimization industry use case examples

BCG estimates the value created, from a subset of quantum optimization use cases, to \$100–220B USD*

 Finance - Portfolio optimization - Underwriting optimization - Anti-money laundering	 Energy and utilities - Grid optimization - Energy trading optimization - Vehicle-to-grid fleet charging	 Health care and life sciences - mRNA optimization - Protein folding - Lead optimization	 Automotive - Vehicle routing - Vehicle design - Production scheduling
 Electronics - Interconnect routing - Infrastructure and network - Production scheduling	 Oil and gas, chemical - Commodity trading - Maritime energy routing - Workforce optimization	 Logistics and telco - Logistics optimization - Network optimization - Routing optimization	 Aerospace and defense - Aircraft design - Satellite mission planning - Flights routing

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* Source: BCG, "The Long-Term Forecast for Quantum Computing Still Looks Bright," 2024.

Vanguard

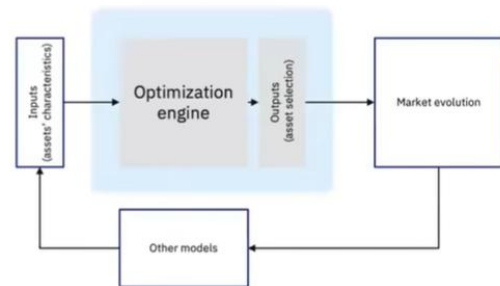
Portfolio optimization for fixed income assets

IBM partnered with Vanguard to conduct research on hybrid quantum algorithms for [optimal portfolio construction](#) in the context of fixed income assets, with the goal of improving the replication quality of selected financial indexes using smaller, less complex portfolios that are cheaper to manage.

The team developed a scalable sampling-based approach, tested up to **109 bonds** on utility-scale hardware using up to **4,200 gates**, producing solutions **within 0.5% of the optimal**.

"The study found that the quantum-classical workflow consistently outperformed a purely classical local search approach, especially as the problem size increased."

"IBM and Vanguard explore quantum optimization for portfolio construction"



The figure above shows the workflow implemented for this solution. The generic portfolio creation problem can be defined as selecting the optimal subset of assets (in the asset universe) that better corresponds to a target set of properties. The asset characteristics (returns, risks, etc.) that are input to the optimization engine are considered as given deterministic inputs, coming from other models.

Read the paper: [arXiv:2508.13557](https://arxiv.org/abs/2508.13557)

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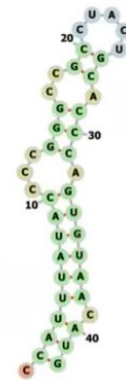
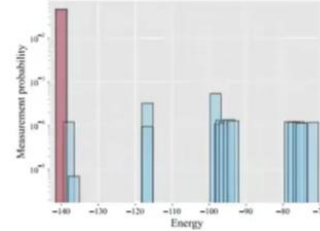
Predicting mRNA secondary structures with quantum optimization techniques

Moderna partnered with IBM to develop quantum computing algorithms for predicting mRNA secondary structures at greater scale and accuracy, with the goal of unveiling novel and more diverse mRNA therapies.

Running the CVaR-VQA algorithm on IBM Quantum Eagle and IBM Quantum Heron yielded accurate insights into folding mechanisms for mRNA sequences of up to 42 nucleotides (mapped to 10–80 qubits) and their associated energy states that match the results of the classical solver CPLEX.

“Our world is a quantum mechanical world, and so when we try to simplify things in a deterministic world, we’re not going to really be able to emulate that complexity.”

Wade Davis
Senior Vice President, Digital for Business, Moderna



These figures show (a) the measurement probability of sampled bitstrings plotted against bitstring energies for the 80-qubit problem, where the red bar indicates the probability corresponding to the lowest energy bitstring; and (b) the optimal folded structure of the 42-nucleotide, 80-qubit mRNA sequence based on the corresponding lowest energy bitstring found by the hardware run.

Read the papers: [arXiv:2405.20328](https://arxiv.org/abs/2405.20328); [arXiv:2507.18817](https://arxiv.org/abs/2507.18817)

Read the case study: <https://ibm.biz/Bdp4gw>

Explore the demo: ibm.biz/BdGAB6

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Takeaways

Today’s quantum computers are tools for algorithmic discovery

We must complement theoretical pen-and-paper research with empirical trial-and-error research to expand our capabilities. The hardware is ready to support this journey, with quantum and classical working synergistically!

Benchmarking is key on the path to quantum advantage

We need standardized, comparable results to identify where quantum algorithms (particularly heuristics) can outperform classical ones—and to discover real-world applications that benefit from them.

We must look beyond the “standard” problems and algorithms.

Less explored domains like multi-objective, robust, stochastic, dynamic optimization may reveal new opportunities for quantum advantage. Let’s revisit them from a new perspective!

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AI for Optimization Example: Non-linear Optimization over Graphs

Goal: Find (near) optimal solutions for large scale problems faster

Given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$,

Node features: $\mathcal{X}_v \in \mathbb{R}^{d_v}$ for $v \in \mathcal{V}$,

Edge features: $\mathcal{X}_e \in \mathbb{R}^{d_e}$ for $e \in \mathcal{E}$

Decision variables: $x_v \in \mathcal{X}_v$ and $x_e \in \mathcal{X}_e$:

$$\min_{x_v, x_e} f_o(x_v, x_e)$$

Subject to:

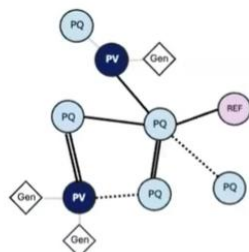
$$g_i(x_v, x_e) = 0 \quad \forall i \in \mathcal{J}_{eq}$$

$$h_j(x_v, x_e) \leq 0 \quad \forall j \in \mathcal{J}_{in}$$



40

Optimal Power Flow (OPF) Problem



- Non-convex network flow problem
- Very challenging to solve to solve numerically – in particular for large-scale networks in near-real time operational settings.
- DoE ARPA-E grid competitions (2018 - 2024).
- Need to find (near) optimal solutions faster.

Economic Dispatch Model

Goal: Find economically optimal (lowest possible cost) output/setpoint of all generators so that all load and physical constraints are met

Gen Cost function for each generator g : $C_g(P_g)$

Objective function: $C(P_g) = \sum_{g \in \mathcal{G}} C_g(P_g)$

Subject to:

Active & Reactive Power Balance (Kirchoff's laws): $P_i^G - P_i^D = \sum_j P_{ij}(V, \theta)$
 $Q_i^G - Q_i^D = \sum_j Q_{ij}(V, \theta)$

Generator Active & Reactive Limits: $P_g^{min} \leq P_g \leq P_g^{max}$
 $Q_g^{min} \leq Q_g \leq Q_g^{max}$

Bus Voltage Limits: $V_i^{min} \leq V_i \leq V_i^{max}$

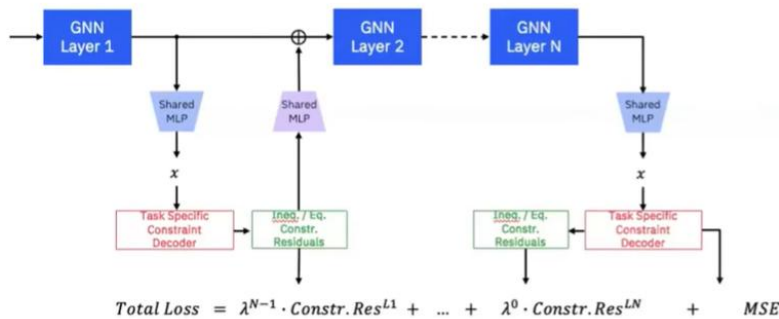
Transmission Line Limits: $|S_{ij}| = \sqrt{P_{ij}^2 + Q_{ij}^2} \leq S_{ij}^{max}$



AI Approach

Geometric Neural Corrective Optimizer

- Heterogeneous Graph Neural Network with novel corrective step



Results achieved to-date

- Model with up to 20M parameters.
- Model trained on extensive synthetic optimal power flow data.
- 35x faster compared to State-of-art classical optimizers (IPOPT)
- Optimality gap <0.2%, feasibility violation <0.5%, compared with state-of-the-art AI-surrogate optimization approaches.
- Benchmarked on realistic instances with several thousand nodes.

Takeaways

Quantum is real and rapidly advancing. Will provide a new computational tool for data science

Both Quantum & AI give opportunity to approach problems in a new way

Benchmarking is key for determining which optimization approach is best